Marine Remote Sensing Object Detection in Complex Backgrounds Based on Diffusion Models and Multidirectional Spatial Attention

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***Abstract*—Rapid red-tide detection and ship target extraction in nearshore remote sensing images are important for marine ecological monitoring and maritime management. However, nearshore images are easily affected by marine weather conditions and sensor imaging processes, often suffering from cloud cover and stripe noise. These interferences introduce distracting pseudofeatures and weaken target–background discriminability, limiting the stability and robustness of existing detectors. Moreover, the limited number of difficult samples under such interference further increases detection difficulty in complex nearshore scenes. To address these issues, we construct the nearshore complex-scene marine remote sensing dataset (NCSMRS), which contains 1369 images covering three typical nearshore scene types: cloud cover, stripe noise, and clean background, with bounding-box annotations for marine targets. We further propose the diffusion-augmented multidirectional attention detection framework (DMA-Det). DMA-Det adopts layout-guided diffusion augmentation to enrich difficult training samples and introduces a semantic-edge quality filter (SEQF) to improve the reliability of synthetic data through semantic consistency validation and edge quality filtering. In addition, a multidirectional spatial attention (MDSA) module is designed in the detection network to model axial and nonaxial spatial dependencies and recalibrate multiscale features, thereby enhancing target-aware feature representations and suppressing interference from complex backgrounds. Extensive experiments on NCSMRS show that DMA-Det outperforms representative detectors and achieves stable performance across different nearshore scenes, with mAP_{50} and $mAP_{50\text{-}95}$ scores of 76.2% and 49.7%, respectively.**

***Index Terms*—Complex nearshore remote sensing scenes, generative data augmentation, object detection, red tide, spatial attention.**

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I. Introduction

NEARSHORE regions are critical zones where coastal human activities, maritime transportation, and marine ecological processes are closely interconnected. High-resolution optical remote sensing satellites enable large-scale and periodic acquisition of nearshore images, providing essential data support for applications such as rapid red-tide detection [1], [2], [3] and ship recognition [4], [5]. However, as shown in Fig. 1, nearshore images are often affected by cloud cover and stripe noise caused by sensor pushbroom imaging [6], [7]. These factors introduce distracting pseudofeatures that weaken the model's ability to distinguish real targets. Meanwhile, red-tide regions often coexist with ships and islands, further complicating target recognition and extraction in complex nearshore scenes. Although existing computer vision-based methods have achieved high accuracy in red-tide detection, they usually rely on ideal, interference-free images and struggle to adapt to complex nearshore scenes with cloud occlusion, stripe noise, and coexisting targets. With the continuous accumulation of satellite images, a fast, efficient, and robust detection framework for such scenes is urgently needed [8], [9]. Therefore, this article formulates red-tide monitoring and marine target extraction in complex nearshore scenes as a unified object detection task. The objective is to rapidly localize suspicious red-tide regions and coexisting marine targets in complex nearshore scenes, rather than to replace fine-grained red-tide boundary mapping or concentration retrieval.

In recent years, deep learning has significantly advanced remote sensing object detection. To address scale variations, dense object distributions, and classification–localization conflicts, existing methods mainly focus on multiscale feature representation, contextual modeling, feature fusion, and task-decoupled optimization [8], [10], [11], [12]. For small objects with weak visual responses, recent studies introduce adaptive receptive fields, dynamic feature learning, local–global feature coordination, and feature alignment fusion to enhance target representations and preserve fine-grained details [9], [13], [14], [15]. To improve robustness in complex or degraded scenes, attention-based enhancement, multimodal fusion, and background suppression have also been explored to alleviate cluttered backgrounds and cloud/mist interference [16], [17], [18].

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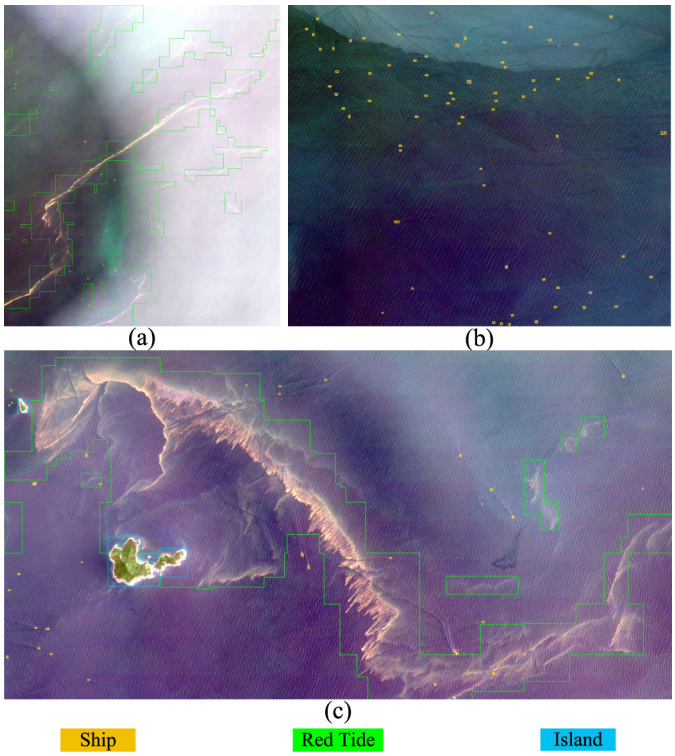


Fig. 1. Representative complex nearshore scenes with object annotations. (a) Cloud-cover overlaps with red-tide regions, introducing strong visual interference. (b) Stripe noise caused by pushbroom imaging affects sea-surface texture features. (c) A large-scale red-tide region exhibits uneven biomass distribution, while thin cloud cover and stripe noise coexist in the scene.

Existing detection models still face special challenges in complex nearshore scenes. On the one hand, high-quality annotated samples in complex nearshore scenes are scarce, and difficult samples under cloud-cover and stripe-noise interference are insufficiently annotated, making it difficult to cover diverse target morphologies and background interference, thereby limiting the robustness and generalization ability of detection models. On the other hand, cloud cover and stripe noise introduce interfering pseudofeatures. These pseudofeatures are similar to the features of targets such as red tide and ships in terms of color, texture, and edge morphology, which can easily cause the model to produce incorrect responses. In particula, cloud cover interferes with the feature representation of targets and increases the difficulty of target recognition and localization. Thin clouds and cloud edges may be misclassified as slight red-tide regions, while isolated small cloud patches can easily be misclassified as ships. Stripe noise disrupts the continuity of sea-surface textures and increases the risk of false detections.

To address the above issues, we construct the nearshore complex-scene marine remote sensing dataset (NCSMRS), which contains 1369 images covering three representative nearshore scenes: cloud cover, stripe-noise interference, and clean background. NCSMRS provides bounding-box annotations for major nearshore targets, including red tide, ships, and islands, enabling object detection research in complex nearshore scenes. The island category is included because islands commonly coexist with red-tide regions and ships and are important components of complex nearshore scenes. Moreover, their bright beaches and reef structures may produce color and brightness responses similar to those of red-tide regions or ships, leading to false detections. Modeling islands as a separate category, therefore, provides a more complete representation of nearshore scenes and helps reduce such confusion. Furthermore, we propose the diffusion-augmented multidirectional attention detection framework (DMA-Det) for object detection in complex nearshore scenes. DMA-Det combines layout-guided diffusion augmentation with a semantic-edge quality filter (SEQF) to generate and filter reliable synthetic remote sensing images, thereby mitigating sample scarcity. Meanwhile, the multidirectional spatial attention (MDSA) module suppresses background interference caused by cloud cover and stripe noise and enhances target-aware feature representations. The main contributions of this article are summarized as follows.

1) We construct the NCSMRS dataset, which covers complex nearshore scenes affected by cloud-cover and stripe-noise interference, with the main targets annotated in each scene.
2) We propose DMA-Det, a DMA-Det that leverages generative data augmentation and MDSA to alleviate sample scarcity and suppress background interference.
3) We design SEQF to filter synthetic samples from two aspects, namely semantic consistency and edge quality, thereby improving the quality of synthetic data. We also design the MDSA module, which suppresses background interference such as cloud cover and stripe noise through an attention reweighting process while modeling multidirectional spatial dependencies to enhance target feature representation.
4) Comprehensive experiments demonstrate that the proposed method outperforms existing state-of-the-art methods, achieving mAP_{50} and $mAP_{50\text{-}95}$ scores of 76.2% and 49.7%, respectively. These results validate its effectiveness in suppressing background interference such as cloud cover and stripe noise in complex nearshore scenes.

The remainder of this article is organized as follows. Section II introduces the related work. Section III describes the proposed DMA-Det. We present the experimental results and analysis in Section IV. Finally, a short conclusion is drawn in Section V.

II. Related Works

A. Red-Tide Remote Sensing Monitoring

Red-tide monitoring is an important coastal remote sensing task [19]. Traditional methods mainly rely on spectral indices, threshold segmentation, or ocean-color products, which are effective for large-scale bloom observation but are sensitive to water-color variations, atmospheric conditions, and sensor differences [20], [21], [22]. With the development of deep learning, pixelwise classification and semantic segmentation models have been used to delineate red-tide regions from satellite images [23], [24]. Recent work has also investigated transformer-based models for fine-grained red-tide segmentation in multispectral imagery [25].

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However, most segmentation-based methods focus on pixel-level red-tide extraction alone [26] while paying less attention to unified target perception in complex nearshore scenes. In real coastal imagery, red-tide regions often coexist with ships, islands, cloud interference, and sensor-induced stripe noise, which makes direct segmentation difficult [27]. With the rapid accumulation of massive remote sensing imagery, efficient multitarget extraction in nearshore scenes has become an urgent demand. Therefore, our work aims to construct a fast and effective object detection framework for complex nearshore scenes. Compared with the pixelwise masks used in semantic or instance segmentation, horizontal bounding boxes cannot precisely delineate the diffuse and irregular boundaries or spatial extent of red-tide regions. Consequently, the annotated regions inevitably include seawater, clouds, and other nontarget pixels, potentially introducing background interference into the learning of target features within the annotated regions. Nevertheless, the bounding-box-based formulation remains appropriate for rapid screening in complex nearshore scenes because it generally requires less annotation effort, supports efficient inference, and provides candidate regions for subsequent visual interpretation or fine-grained analysis.

B. Generative Data Augmentation

Compared with conventional augmentation based on geometric or photometric transforms, diffusion-based synthesis [28], [29], [30] supports finer control via text, category labels, visual priors, or spatial layouts. For natural-image detection, InstaGen [31] augments diffusion generation with an instance-level grounding head so that images and bounding-box labels are jointly produced. Fang et al. [29] combine controllable diffusion, visual priors, and CLIP-based filtering to obtain detection-oriented training pairs. MPAD [32] targets few-shot regimes through multiperspective synthetic augmentation, whereas RegionDiffusion [33] performs region-level controllable generation to improve label fidelity and sample quality.

In remote sensing, AeroGen [30] couples layout conditioning with diffusion for detection-centric data synthesis. LSDGen [34] further injects intercategory relational priors to obtain plausible layouts and refines synthetic sets through filtering. RSGen [35] adds edge guidance to sharpen instance boundaries in layout-to-image generation. Zhang et al. [36] model object-background coupling in a cascaded autoregressive diffusion design, while CC-Diff++ [37] stresses foreground–background coherence in spatially controllable text-to-image synthesis. TerraGen [38] generalizes the idea to a unified multitask layout generation framework for remote sensing augmentation. Different from these studies, our work shifts the focus from natural images and land-dominated remote sensing scenes to marine scenes affected by cloud interference and stripe noise, where irregular red-tide regions coexist with small ships and islands.

C. Complex Remote Sensing Object Detection

Since remote sensing object detection often relies on general detection frameworks, recent generic detectors provide important foundations for complex remote sensing scenes. Recent YOLO variants improve the accuracy–efficiency tradeoff through NMS-free training and attention-centric architecture design [39], [40]. Mamba YOLO [41] further introduces state-space modeling into YOLO-style detection to achieve efficient global feature representation. For DETR-based detectors, RT-DETR [42] extends DETR-style models to real-time detection with efficient encoding and query selection. Recent DETR variants further improve query filtering, proposal refinement, and training efficiency [43], [44], [45]. More recently, DEIMv2 [46] further enhances real-time detection through DINOv3-based feature adaptation. However, remote sensing images still suffer from scale variation, dense distributions, weak small-object features, and complex background interference [47], [48].

For remote sensing object detection, existing studies have improved robustness mainly through contextual modeling, multiscale representation, and background suppression. CAD-Net [49] enhances feature discrimination by exploiting contextual information and attention mechanisms. PCViT [8] introduces pyramid transformer modeling to capture multiscale contextual representations. To cope with scale variation and weak small-object responses, RFHA-YOLO [9] and M2FE-YOLO [11] strengthen multiscale representation and small-object feature enhancement in cluttered scenes. For complex or degraded scenes, CM-YOLO [18] suppresses background interference in cloud- and mist-contaminated images, while Xu et al. [50] address tiny, arbitrary-oriented ship detection under dense distributions, large aspect ratios, and strong background noise. However, existing methods still show limited adaptability to complex nearshore scenes. Cloud cover and stripe noise introduce interfering pseudofeatures, weakening target–background discriminability and increasing false positives and missed detections for red-tide regions and small ships. Moreover, fuzzy boundaries, extreme aspect ratios, and weak responses in slight red-tide areas make it difficult to accurately capture the complete spatial extent of red-tide regions.

III. Method

In this section, we first describe the data preprocessing procedure in Section III-A, which converts the original multispectral satellite imagery into model-ready image patches. As shown in Fig. 2, the two key components of DMA-Det are then introduced: 1) the data augmentation pipeline in Section III-B, which generates and filters synthetic samples and 2) the object detection model in Section III-C, where the proposed MDSA module is described in detail to suppress background interference and enhance target-aware feature representation.

A. Data Preprocessing

The original satellite imagery contains eight spectral bands with a spatial resolution of 3 m per pixel and includes scenes affected by typical optical degradations, such as cloud cover and sensor stripe noise. During data preprocessing, the RGB bands were first extracted from the original multispectral imagery to obtain standard three-channel RGB images.

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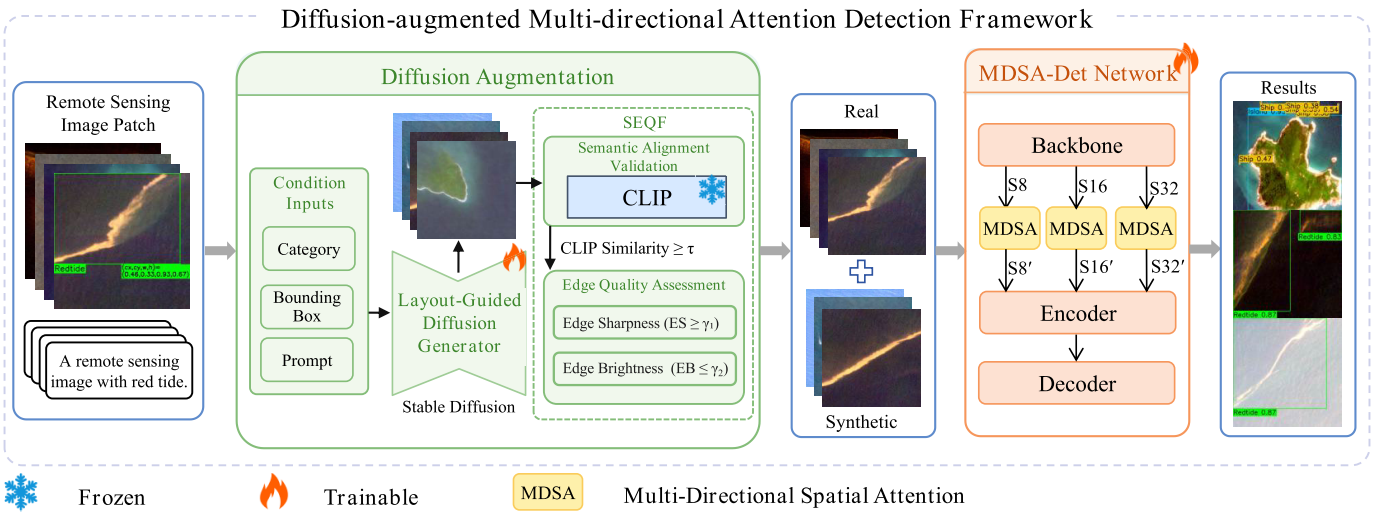


Fig. 2. Overview of the proposed DMA-Det. Quality-assured diffusion augmentation enriches training samples, while the MDSA-enhanced detector suppresses cloud and stripe-noise interference for robust target perception.

After that, invalid edge regions were removed, and the high-resolution images were divided into nonoverlapping patches of 1024 × 1024 pixels. The cropped patches were further screened to remove pure-background images and extremely degraded images in which reliable target annotation was difficult. The remaining image patches were manually annotated with horizontal bounding boxes for three target categories, including red tide, ship, and island. For red-tide annotation, each visually discernible region, including slight red-tide areas, was enclosed by a horizontal bounding box according to its visible extent. Filamentary or spatially continuous regions were treated as one instance, while small fragments within a larger-region box were not annotated separately. Under cloud cover, red tide was annotated where its visual features remained discernible; fully obscured areas were excluded, and cloud-separated portions of a continuous red-tide region were treated as separate instances.

B. Diffusion Augmentation

1) Layout-Guided Diffusion Generation: In the data augmentation stage, inspired by the layout-conditional generation strategy of AeroGen [30], we design a layout-guided diffusion augmentation module to synthesize pseudoremote sensing images. The module is fine-tuned using data from the detector training split of NCSMRS and generates synthetic samples under the joint guidance of text prompts, category labels, and spatial layouts.

During fine-tuning, the conditional diffusion model learns the following mapping:

$$\{T, C_{\text{real}}, B_{\text{real}}\} \rightarrow I_{\text{real}} \tag{1}$$

where T is automatically constructed from the category labels and bounding-box layouts, and I_{real}, C_{real}, and B_{real} denote the real remote sensing images, category labels, and bounding-box layouts, respectively. Each bounding box is represented as (c_x, c_y, w, h), where (c_x, c_y) denotes the box center and (w, h) denotes its width and height.

At inference, pseudoremote sensing images are generated by

$$\{T, C_{\text{syn}}, B_{\text{syn}}\} \rightarrow I_{\text{syn}} \tag{2}$$

where I_{syn}, C_{syn}, and B_{syn} denote the synthetic remote sensing images, category labels, and bounding-box layouts, respectively. Since C_{syn} and B_{syn} serve as generation conditions, the annotations of I_{syn} can be directly derived from them without additional manual labeling.

2) Semantic-Edge Quality Filter: Before detector training, SEQF further filters the synthetic images $\mathcal{I}_{\text{syn}}$ to remove low-fidelity samples. SEQF follows a two-stage filtering pipeline, where semantic alignment validation (SAV) first removes samples with category-level semantic inconsistency, and edge quality assessment (EQA) then filters samples with unreliable edge structures. The retained synthetic samples are formulated as

$$\mathcal{I}^{*}_{\text{syn}} = \left\{ I_{\text{syn}} \mid \text{SAV}\left(I_{\text{syn}}, \mathcal{C}_{\text{syn}}\right) = 1, \text{EQA}\left(I_{\text{syn}}\right) = 1 \right\}. \tag{3}$$

The retained synthetic samples $\mathcal{I}^{*}_{\text{syn}}$ are then combined with real samples for subsequent detector training. Fig. 3 provides a visual comparison between real remote sensing patches and synthetic samples retained after SEQF filtering.

a) SAV: SAV checks whether the generated image I_{syn} is semantically consistent with its specified category labels $\mathcal{C}_{\text{syn}}$. For each category $c \in \mathcal{C}_{\text{syn}}$, a textual prompt T_c is formed and the CLIP similarity between I_{syn} and T_c is computed

$$S_c = \frac{f_{\text{img}}\left(I_{\text{syn}}\right) \cdot f_{\text{text}}\left(T_c\right)}{\left\| f_{\text{img}}\left(I_{\text{syn}}\right) \right\|_2 \left\| f_{\text{text}}\left(T_c\right) \right\|_2} \tag{4}$$

where $f_{\text{img}}(\cdot)$ and $f_{\text{text}}(\cdot)$ denote the CLIP image and text encoders, respectively. The generated image passes SAV only when the similarity score for every specified category meets or exceeds its category-specific threshold

$$\text{SAV}\left(I_{\text{syn}}, \mathcal{C}_{\text{syn}}\right) = \begin{cases} 1, & S_c \geq \tau_c, \ \forall\, c \in \mathcal{C}_{\text{syn}} \\ 0, & \text{otherwise} \end{cases} \tag{5}$$

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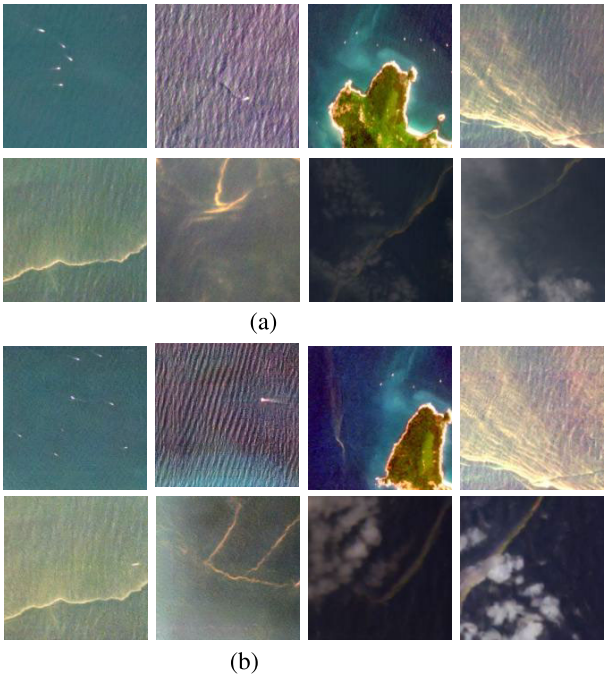


Fig. 3. Visual comparison of real versus synthetic images. (a) Patches cropped from real remote sensing images. (b) Synthetic patches generated directly at patch resolution.

where τ_c denotes the validation threshold for category c, and $\mathcal{C}_{\text{syn}} \subseteq \{\text{Red Tide, Ship, Island}\}$.

b) EQA: EQA evaluates whether the generated image preserves reliable boundary structures. Given a synthesized image I_{syn}, EQA measures edge sharpness and abnormal edge brightness.

1) *Edge Sharpness:* We convert I_{syn} into a grayscale image G and compute the variance of its Laplacian response

$$\text{ES} = \text{Var}\left(\nabla^2 G\right) = \frac{1}{HW}\sum_{i,j}\left(\nabla^2 G(i,j) - \overline{\nabla^2 G}\right)^2 \quad (6)$$

where ∇^2 denotes the Laplacian operator, and higher ES indicates sharper boundary details.

2) *Edge Brightness:* A candidate region mask M is obtained from I_{syn} via HSV color thresholding. Morphological dilation produces a thick outer ring around M

$$\mathcal{E} = \text{Dilate}(M) \setminus M. \quad (7)$$

Let $V(p)$ denote the HSV value channel at pixel p of I_{syn}. Band intensities are collected as $\mathcal{V} = \{V(p) \mid p \in \mathcal{E}\}$. Bright-band pixels form the subset $\mathcal{B} \subseteq \mathcal{E}$

$$\mathcal{B} = \{\, p \in \mathcal{E} : V(p) \geq v_{\text{thr}} \,\}. \quad (8)$$

The edge-brightness statistic is computed as

$$\text{EB} = \begin{cases} \lambda P_q(\mathcal{V}) + (1-\lambda)\dfrac{1}{|\mathcal{B}|}\displaystyle\sum_{p\in\mathcal{B}} V(p), & |\mathcal{B}| \geq n_{\min} \\ P_q(\mathcal{V}), & \text{otherwise} \end{cases} \quad (9)$$

where $P_q(\mathcal{V})$ is the qth percentile of $\mathcal{V}$. The EB parameters are empirically set to $q = 93$, $v_{\text{thr}} = 200$, $n_{\min} = 8$, and $\lambda = 0.65$; v_{thr} selects bright-band pixels, $n_{\min}$ specifies the minimum number of such pixels required to include their mean brightness, and λ balances the percentile and mean brightness terms. When $|\mathcal{B}| < n_{\min}$, EB reduces to $P_q(\mathcal{V})$. A larger EB indicates stronger halos or overbright rims.

A generated image passes EQA only when

$$\text{EQA}\left(I_{\text{syn}}\right) = \begin{cases} 1, & \text{ES} \geq \gamma_1,\ \text{EB} \leq \gamma_2 \\ 0, & \text{otherwise} \end{cases} \quad (10)$$

where γ_1 and γ_2 are experimentally determined thresholds. Algorithm 1 summarizes the complete filtering procedure of SEQF.

Algorithm 1 SEQF

Input: Synthetic image set $\mathcal{I}_{\text{syn}}$, category labels $\mathcal{C}_{\text{syn}}$, prompts $\mathcal{T}$, semantic thresholds $\{\tau_c\}$, and edge-quality thresholds γ_1, γ_2
Output: Retained synthetic image set $\mathcal{I}^*_{\text{syn}}$

```
I*_syn ← ∅
for each I_syn ∈ I_syn do
    Obtain its category set C_syn and corresponding prompts {T_c | c ∈ C_syn}
    for each c ∈ C_syn do
        Compute the CLIP similarity S_c by Eq. (4)
    end for
    if S_c < τ_c for any c ∈ C_syn then
        continue
    end if
    Compute ES and EB by Eqs. (6)–(9)
    if ES ≥ γ1 and EB ≤ γ2 then
        I*_syn ← I*_syn ∪ {I_syn}
    end if
end for
return I*_syn
```

C. MDSA-Det Network

The detector is trained using both real remote sensing patches and retained synthetic samples. It consists of a backbone, an encoder, and a decoder. The backbone extracts multiscale feature maps, and the proposed MDSA module recalibrates them by capturing horizontal, vertical, and diagonal dependencies. The recalibrated features are then fed into the encoder for cross-scale feature fusion, and the decoder predicts object categories and bounding-box coordinates.

1) Multidirectional Spatial Attention: In complex nearshore scenes, cloud cover and stripe noise introduce interfering pseudofeatures that weaken target–background discriminability. Red-tide detection is further challenged by fuzzy boundaries, irregular shapes, and large aspect-ratio variations. To suppress such background interference and enhance target-aware representation, we introduce the MDSA module. As shown in Fig. 4, MDSA is applied to the multiscale backbone feature maps from S_8, S_{16}, and S_{32}. For each scale, a structurally identical MDSA module with unshared parameters is used to perform scale-specific spatial recalibration. Given an input feature map $F_{\text{in}} \in \mathbb{R}^{C\times H\times W}$, each module generates attention weights to reweight target-relevant responses.

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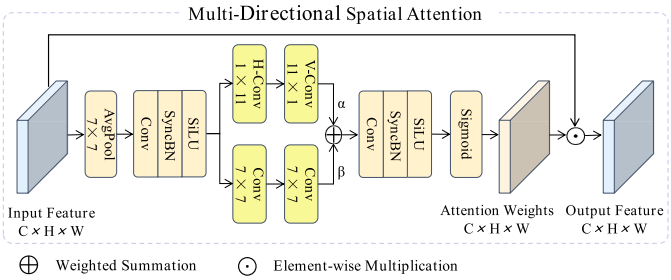


Fig. 4. Architecture of the proposed MDSA module. MDSA captures axial and nonaxial spatial dependencies through convolution branches, fuses them by weighted summation, and generates attention weights to recalibrate the input feature map.

Initially, local contextual information is aggregated through average pooling with a window size of 7×7, followed by 1×1 convolution, synchronized batch normalization (SyncBN), and sigmoid linear unit (SiLU) activation to obtain F_{base}

$$F_{\text{base}} = \text{SiLU}\left(\text{SyncBN}\left(\text{Conv}_{1\times1}\left(\text{AvgPool}_{7\times7}\left(F_{\text{in}}\right)\right)\right)\right). \quad (11)$$

Based on F_{base}, MDSA uses two complementary branches to capture directional spatial dependencies. The axial branch applies cascaded horizontal and vertical strip convolutions to model long-range dependencies along coordinate axes

$$F_{\text{HV}} = \text{VConv}_{11\times1}\left(\text{HConv}_{1\times11}\left(F_{\text{base}}\right)\right). \quad (12)$$

The nonaxial branch uses two consecutive 7×7 convolutions to capture broader spatial context and nonaxis-aligned patterns

$$F_{\text{NA}} = \text{Conv}^{2}_{7\times7}\left(\text{Conv}^{1}_{7\times7}\left(F_{\text{base}}\right)\right). \quad (13)$$

The outputs of the two branches are fused to integrate axial and nonaxial spatial cues

$$F_{\text{fuse}} = \alpha F_{\text{HV}} + \beta F_{\text{NA}} \quad (14)$$

where α and β denote the fusion weights of the axial and nonaxial branches, respectively, with $\alpha + \beta = 1$. The fused features are further transformed by a 1×1 convolution, SyncBN, SiLU activation, and sigmoid activation to generate the attention weight map A

$$A = \sigma\left(\text{SiLU}\left(\text{SyncBN}\left(\text{Conv}_{1\times1}\left(F_{\text{fuse}}\right)\right)\right)\right) \quad (15)$$

where $\sigma(\cdot)$ denotes the sigmoid function. The attention map A adaptively emphasizes target-relevant regions while suppressing responses from cloud cover, stripe noise, and other cluttered nearshore backgrounds. Finally, the input feature is recalibrated as

$$F_{\text{out}} = F_{\text{in}} \odot A. \quad (16)$$

For each scale S_8, S_{16}, and S_{32}, the recalibrated feature F_{out} is forwarded to the encoder for multiscale feature fusion, followed by the decoder for category prediction and bounding-box regression.

IV. Experiments

A. Dataset

Original images were acquired from the SuperDove multispectral sensor aboard PlanetScope satellites in April 2022, covering the offshore coastal regions of Guangdong Province, China. A summary of the PlanetScope scene IDs and acquisition dates can be found in Table I. The NCSMRS was generated after performing the extraction of the three primary color channels, image cropping, data cleaning, and expert visual interpretation. To characterize the complexity of nearshore imaging conditions, we annotated the images using three scene-type labels: cloud cover, stripe noise, and clean background. Detailed categorywise and scenewise statistics are presented in Table II. The categorywise image counts are based on image-level occurrence, and one image may contain multiple target categories. For scene types, cloud cover and stripe noise are not mutually exclusive because both degradations may appear in the same image, whereas clean background denotes images without visually obvious cloud-cover or stripe-noise interference.

For detector evaluation, seven source scenes were reserved as an independent test set and cropped into nonoverlapping patches, yielding test patches that accounted for approximately 15% of all annotated patches. The remaining 12 scenes were also cropped into nonoverlapping patches, and their patches were split into detector training and validation sets at an 80%/20% ratio, yielding overall proportions of approximately 68%, 17%, and 15% for training, validation, and test, respectively. The detector training set was further divided into diffusion-model training and validation sets at an 80%/20% ratio. In addition, as summarized in Table III, after SEQF filtering, diffusion-based augmentation adds approximately 200 synthetic training instances for red tide and ship, and 109 instances for island to the detector training split. These synthetic samples are provided with bounding-box annotations derived from the same layout conditioning used at generation time, yielding a moderate expansion of supervision without dominating the real data.

B. Experimental Setup

1) Implementation Details: The diffusion model is fine-tuned for 400 epochs using the AdamW optimizer with a learning rate of 1×10^{-5}. At inference, sampling is performed with 50 denoising steps, and η is set to 0.2. For SAV, the category-specific thresholds are set to $\tau_{\text{ship}} = 0.35$, $\tau_{\text{red tide}} = 0.29$, and $\tau_{\text{island}} = 0.30$. For EQA, the thresholds are set to $\gamma_1 = 15.31$ and $\gamma_2 = 253.68$. For the object detection experiments, the comparison models use their optimal configurations established on the COCO dataset. To ensure convergence, S-DETR, R-DETR, and PCViT are trained for 24 epochs, YOLO-based models are trained for 200 epochs, and DEIMv2-S and the proposed DMA-Det are both trained for 132 epochs. DMA-Det is optimized using AdamW with a base learning rate of 5×10^{-4}, a weight decay of 1×10^{-4}, and betas of $(0.9, 0.99)$. For the proposed MDSA module, a module-specific learning rate of 4×10^{-3} and a weight decay of 1×10^{-4} are used. The fusion weights α and β are set

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TABLE I

OVERVIEW OF SATELLITE IMAGERY USED IN THE DATASET. SCENE IDS FOLLOW THE PLANETSCOPE NAMING CONVENTION; ALL SOURCE FILES ARE _3B_ANALYTICMS_SR_8B_HARMONIZED[_CLIP].TIF. SCENE IDS IN BOLD DENOTE THE SOURCE SCENES ASSIGNED TO THE INDEPENDENT TEST SET

Satellite	Resolution	Date	Scene ID		
PlanetScope	3 m/px	2022-04-06	20220406_021018_66_2420 **20220406_022443_59_225a**	20220406_021020_95_2420 20220406_022445_87_225a	20220406_021141_72_2449
		2022-04-07	**20220407_021826_71_2458** 20220407_025646_56_2416	20220407_021829_00_2458	**20220407_025644_26_2416**
		2022-04-08	20220408_021203_16_2427	**20220408_021313_18_2430**	
		2022-04-09	**20220409_020740_54_2455**	**20220409_024515_29_249b**	20220409_021234_95_2429
		2022-04-10	20220410_021109_59_241f	20220410_023951_12_248b	
		2022-04-12	20220412_024819_97_249e		
		2022-04-13	**20220413_022048_28_2262**		
		2022-04-14	20220414_024142_99_247c		

TABLE II

DETAILED STATISTICS OF THE NCSMRS DATASET UNDER DIFFERENT NEARSHORE SCENE TYPES. THE TOTAL COLUMN REPORTS UNIQUE IMAGES; OVERLAPPING SCENES AND TARGET CATEGORIES MAY RESULT IN TOTALS SMALLER THAN DIRECT SUMS

Scene type	Count type	Red Tide	Ship	Island	Total
Cloud cover	Images	414	273	22	620
	BBoxes	644	518	25	1187
Stripe noise	Images	120	188	80	322
	BBoxes	185	303	101	589
Clean background	Images	206	325	221	594
	BBoxes	325	891	256	1472
All scenes	Images	671	675	310	1369
	BBoxes	1046	1516	368	2930

TABLE III

STATISTICS OF RETAINED SYNTHETIC SAMPLES UNDER DIFFERENT NEARSHORE SCENE TYPES. THE TOTAL COLUMN REPORTS UNIQUE IMAGES; OVERLAPPING SCENES AND TARGET CATEGORIES MAY RESULT IN TOTALS SMALLER THAN DIRECT SUMS

Scene type	Count type	Red Tide	Ship	Island	Total
Cloud cover	Images	72	29	0	91
	BBoxes	88	32	0	120
Stripe noise	Images	19	75	50	128
	BBoxes	23	96	50	169
Clean background	Images	111	65	59	207
	BBoxes	117	112	59	288
All scenes	Images	198	151	109	405
	BBoxes	222	221	109	552

to 0.7 and 0.3. All experiments are implemented based on the PyTorch framework and conducted on a single NVIDIA A800 GPU.

2) Evaluation Metrics: Following the COCO-style object detection evaluation protocol [51], AP_{50} denotes the average precision at an intersection over union (IoU) threshold of 0.5 for each category. mAP_{50} is the mean of AP_{50} over all categories, while $mAP_{50\text{-}95}$ is the mAP averaged over IoU thresholds from 0.50 to 0.95 with a step size of 0.05. AR denotes the class-averaged recall at an IoU threshold of 0.5 with a maximum of 100 detections per image. All detection scores are reported in percent (%).

C. Experimental Results

1) Comparison With State-of-the-Art Methods: **Quantitative comparison.** To validate the effectiveness of the proposed DMA-Det, we compare it with a diverse set of representative detectors on the NCSMRS dataset, as reported in Table IV. At the category level, DMA-Det achieves AP_{50} scores of 55.8%, 83.9%, and 89.0% for red tide, ship, and island, respectively, improving over DEIMv2-S by 1.8, 4.4, and 0.7 percentage points. Overall, DMA-Det shows the strongest overall performance among the compared methods, with AR, mAP_{50}, and $mAP_{50\text{-}95}$ scores of 97.2%, 76.2%, and 49.7%, respectively. Relative to DEIMv2-S, mAP_{50} increases from 73.9% to 76.2%, and $mAP_{50\text{-}95}$ from 48.0% to 49.7%. YOLO-based detectors show limited performance on red-tide detection, where fuzzy boundaries and irregular shapes make accurate localization difficult. RT-DETR-based models achieve relatively competitive overall accuracy, but require substantially larger parameter sizes. These results indicate that SEQF-filtered diffusion augmentation enriches reliable training samples, while MDSA enhances target-aware feature representation through multidirectional spatial modeling, improving detection robustness in complex nearshore backgrounds. Notably, DMA-Det has 9.69 M parameters, comparable to DEIMv2-S with 9.71 M, suggesting that the gains mainly come from the proposed augmentation and attention design rather than model scaling.

Qualitative Comparison: Qualitative results on the NCSMRS dataset are shown in Fig. 5, where DMA-Det is compared with representative detectors. In column Fig. 5(a), DMA-Det suppresses the interference of fragmented clouds on red-tide detection and better preserves the spatial continuity of red-tide regions with uneven biomass distribution. It also alleviates the impact of stripe noise on ship recognition and

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TABLE IV

DETECTION PERFORMANCE COMPARISON ON THE NCSMRS DATASET. THE PARAMETER COUNT REPORTED FOR DMA-DET INCLUDES ONLY THE DETECTION NETWORK. BOLD AND UNDERLINE INDICATE THE BEST AND SECOND-BEST RESULTS IN EACH COLUMN, RESPECTIVELY

Method	Backbone	Params (M)	AR	AP_{50}			mAP_{50}	$mAP_{50\text{-}95}$
				Red Tide	Ship	Island		
YOLOv10-L	CSPDarknet	31.67	59.9	36.0	76.3	80.6	64.3	37.4
YOLOv11-L	CSPDarknet	56.87	58.2	35.6	76.2	81.2	64.3	38.5
YOLOv12-L	CSPDarknet	59.12	66.4	42.6	76.7	67.3	62.2	37.3
YOLOv26-L	CSPDarknet	58.82	60.7	38.4	77.2	64.6	60.1	35.1
MS-DETR	ResNet50	53.49	88.6	25.5	39.4	57.0	40.6	24.7
HPR	Swin-L	307.06	88.9	26.1	68.0	73.0	55.7	31.3
S-DETR	ResNet50	235.14	90.7	52.7	70.6	81.0	68.1	40.9
R-DETR	ResNet50	217.01	88.3	50.1	76.4	77.8	68.1	40.1
RT-DETRv1-X	ResNet101	76.64	64.7	43.8	80.3	81.7	68.6	43.6
RT-DETRv2-X	ResNet101	76.37	94.4	48.7	75.4	83.2	69.1	43.8
PCViT	CNN + ViT	109.73	81.9	44.9	77.4	70.8	64.4	30.6
DEIMv2-S	ViT-T + STA	9.71	96.9	54.0	79.5	88.3	73.9	48.0
DMA-Det (ours)	ViT-T + STA	9.69	**97.2**	**55.8**	**83.9**	**89.0**	**76.2**	**49.7**

TABLE V

DETECTION PERFORMANCE COMPARISON UNDER DIFFERENT NEARSHORE SCENE TYPES ON THE NCSMRS DATASET. BOLD AND UNDERLINE INDICATE THE BEST AND SECOND-BEST RESULTS IN EACH COLUMN, RESPECTIVELY

Method	Cloud cover			Stripe noise			Clean background		
	AR	mAP_{50}	$mAP_{50\text{-}95}$	AR	mAP_{50}	$mAP_{50\text{-}95}$	AR	mAP_{50}	$mAP_{50\text{-}95}$
YOLOv10-L	51.9	56.0	29.7	59.2	65.2	36.6	74.0	79.0	46.9
YOLOv11-L	55.1	58.5	31.3	60.6	62.5	35.3	70.8	77.9	49.9
YOLOv12-L	56.6	60.8	34.1	63.9	55.0	29.9	78.9	78.2	50.8
YOLOv26-L	51.6	51.9	28.4	64.4	55.7	31.5	70.5	75.2	45.5
MS-DETR	84.2	33.9	18.5	89.4	47.9	30.8	95.0	48.7	30.5
HPR	84.5	52.2	29.7	89.4	62.6	38.2	94.6	60.9	32.2
S-DETR	87.4	63.0	36.3	92.4	72.2	46.0	93.2	75.4	45.8
R-DETR	85.7	62.8	35.1	90.2	72.6	43.3	93.0	74.9	44.2
RT-DETRv1-X	63.3	63.3	39.1	65.3	72.6	48.6	71.7	75.8	48.3
RT-DETRv2-X	93.2	64.9	41.0	94.9	73.1	49.4	95.6	75.3	45.5
PCViT	78.8	57.9	26.4	84.0	75.3	38.1	88.4	74.9	35.1
DEIMv2-S	96.0	68.7	43.5	**98.1**	77.7	51.6	98.7	78.8	51.2
DMA-Det (ours)	**96.2**	**71.7**	**45.5**	98.0	**81.3**	**54.7**	**99.1**	**81.8**	**52.8**

produces fewer false positives for ships. In column Fig. 5(b), small ships are covered by thin clouds or surrounded by cloud regions, making them prone to missed detections and false positives. DMA-Det suppresses cloud interference effectively and achieves more stable ship detection. In column Fig. 5(c), red-tide regions under thicker cloud cover exhibit weak responses, while cloud-edge structures may cause false positives. Compared with YOLOv26-L, RT-DETRv2-X, and DEIMv2-S, DMA-Det maintains more consistent red-tide localization with fewer spurious detections near cloud edges. Overall, DMA-Det produces predictions that are more consistent with the ground truth and effectively reduces false positives and false negatives under complex nearshore backgrounds.

2) Scenewise Performance: Table V reports the results of the same panel of detectors under three representative nearshore scene types: cloud cover, stripe-noise interference, and relatively clean sea surface. Compared with the clean-background subset, the cloud-cover subset generally leads to lower detection accuracy, indicating that cloud occlusion poses a challenge to nearshore object detection. For stripe-noise scenes, most detectors also show degraded performance. Across all three scene types, DMA-Det achieves the highest mAP_{50} and $mAP_{50\text{-}95}$ scores, while its AR also ranks first or second across the three subsets. These results demonstrate the robust scenewise performance of DMA-Det under different nearshore imaging conditions.

3) Generality of Diffusion Augmentation: To further examine the generality of the proposed diffusion augmentation strategy, we apply the same augmentation pipeline to representative detectors. As presented in Table VI, most detectors benefit from the augmented samples. For the accuracy metrics, diffusion augmentation improves 11 of the 12 detectors for each of mAP_{50} and $mAP_{50\text{-}95}$, with larger gains on MS-DETR, YOLOv12-L, and YOLOv26-L. For AR, most detectors also show positive changes, although slight decreases are observed for HPR, S-DETR, R-DETR, and RT-DETRv2-X. Overall, although the magnitude of the gains varies across detectors, the proposed diffusion augmentation strategy improves the mAP metrics of most baseline detectors. These results suggest that its effectiveness is somewhat architecture-dependent. Nevertheless, the strategy is not limited to DMA-Det and can provide useful additional supervision for different detector

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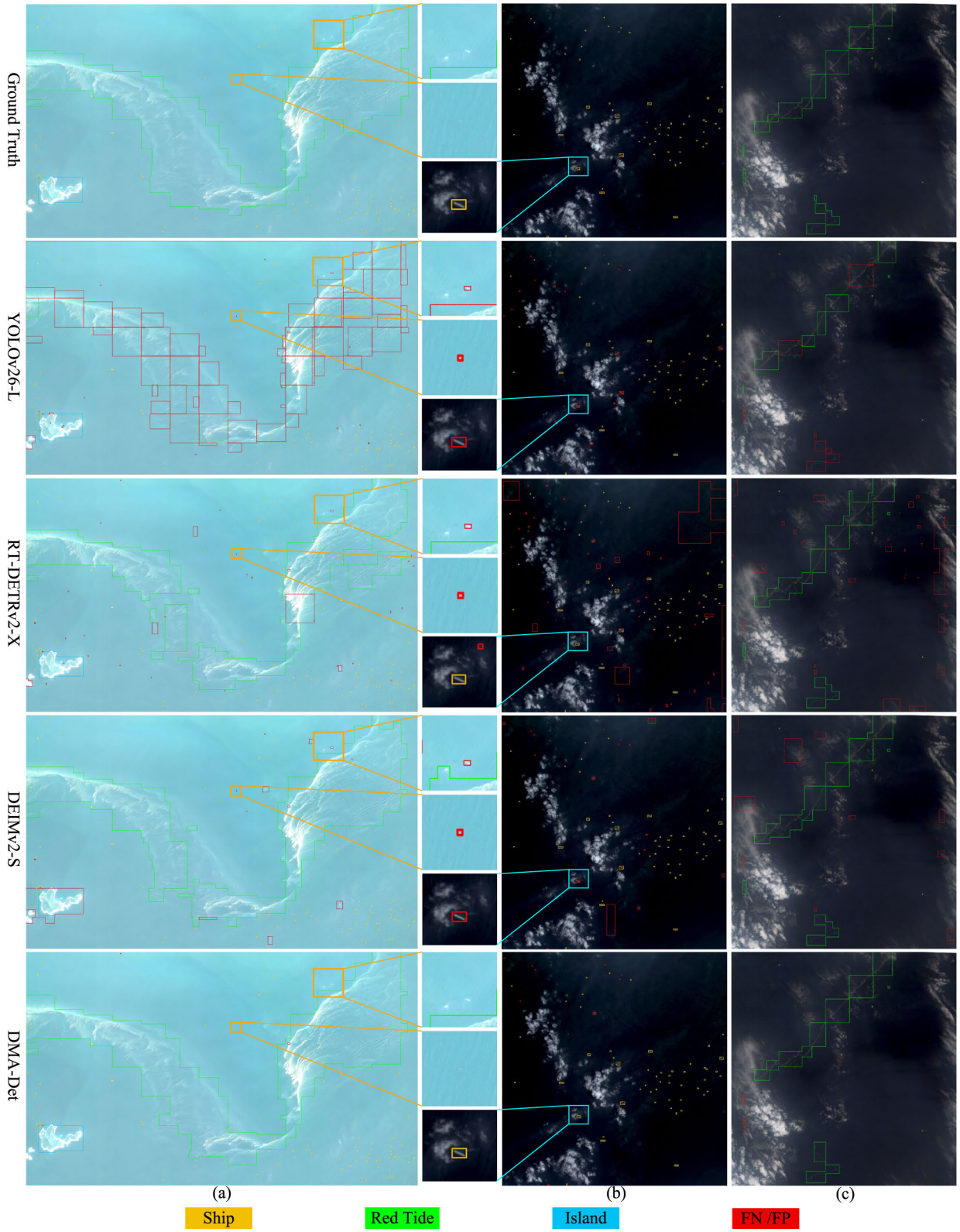


Fig. 5. Qualitative comparison on the NCSMRS dataset between DMA-Det and mainstream representative methods. (a)–(c) Present three representative complex nearshore scenes. (a) Mixed cloud-cover and stripe-noise scene in which fragmented clouds overlap red tide regions with uneven biomass distribution and stripe noise affects sea-surface texture features. (b) Cloud-cover scene in which thin clouds partially obscure or surround small ships. (c) Cloud-cover scene in which thick clouds obscure red tide regions and cloud-edge structures resemble red tide patterns. The detection confidence threshold is fixed at 0.3. The bottom legend explains the class colors; red boxes mark false negatives or false positives. For overlapping, edge touching, and extremely close detections, we merged the corresponding boxes to improve visual clarity; as a result, some bounding boxes in the figure may appear irregular in shape.

families, suggesting a degree of applicability across detector architectures.

4) Evaluation on the GF-1 WFV Dataset: To further evaluate DMA-Det on remote sensing imagery, we construct an

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TABLE VI

EFFECT OF DIFFUSION AUGMENTATION ON REPRESENTATIVE BASELINE DETECTORS. "W/O AUG." DENOTES THE ORIGINAL RESULTS WITHOUT DIFFUSION-AUGMENTED SAMPLES, WHILE "W/AUG." DENOTES THE RESULTS OBTAINED BY APPLYING THE SAME DIFFUSION AUGMENTATION STRATEGY. Δ INDICATES THE PERFORMANCE CHANGE IN PERCENTAGE POINTS

Method	AR			mAP_{50}			mAP_{50-95}		
	w/o Aug.	w/ Aug.	Δ	w/o Aug.	w/ Aug.	Δ	w/o Aug.	w/ Aug.	Δ
YOLOv10-L	59.9	63.9	+4.0	64.3	67.1	+2.8	37.4	39.7	+2.3
YOLOv11-L	58.2	63.2	+5.0	64.3	65.9	+1.6	38.5	38.4	-0.1
YOLOv12-L	66.4	66.5	+0.1	62.2	68.7	+6.5	37.3	43.0	+5.7
YOLOv26-L	60.7	65.7	+5.0	60.1	64.5	+4.4	35.1	38.9	+3.8
MS-DETR	88.6	89.1	+0.5	40.6	54.1	+13.5	24.7	33.2	+8.5
HPR	88.9	88.0	-0.9	55.7	57.9	+2.2	31.3	32.7	+1.4
S-DETR	90.7	90.3	-0.4	68.1	71.5	+3.4	40.9	44.4	+3.5
R-DETR	88.3	87.8	-0.5	68.1	64.1	-4.0	40.1	40.8	+0.7
RT-DETRv1-X	64.7	65.7	+1.0	68.6	70.8	+2.2	43.6	46.1	+2.5
RT-DETRv2-X	94.4	94.1	-0.3	69.1	71.3	+2.2	43.8	45.5	+1.7
PCViT	81.9	82.9	+1.0	64.4	65.8	+1.4	30.6	33.3	+2.7
DEIMv2-S	96.9	97.0	+0.1	73.9	75.4	+1.5	48.0	49.0	+1.0

TABLE VII

DETECTION PERFORMANCE COMPARISON ON THE ADDITIONAL EXTERNAL DATASET. BOLD AND UNDERLINE INDICATE THE BEST AND SECOND-BEST RESULTS IN EACH COLUMN, RESPECTIVELY

Method	AR	AP_{50}			mAP_{50}	mAP_{50-95}
		Ship	Island	Red Tide		
YOLOv10-X	46.9	43.6	64.8	25.4	44.6	22.5
YOLOv11-X	49.8	46.2	77.0	24.0	49.1	24.5
YOLOv12-X	36.2	17.1	76.0	34.8	42.7	24.6
YOLOv26-X	55.1	**49.3**	75.4	39.1	54.6	31.1
MS-DETR	71.1	24.2	76.0	39.3	46.5	28.5
HPR	**83.0**	33.2	79.0	35.5	49.3	28.5
S-DETR	73.5	28.8	78.9	38.7	48.8	29.9
R-DETR	59.1	32.8	67.8	**44.5**	48.4	28.8
RT-DETRv1-X	49.3	38.9	74.5	32.4	48.6	28.5
RT-DETRv2-X	77.4	41.2	79.2	33.6	51.3	29.1
PCViT	66.7	23.4	64.5	14.5	34.2	18.4
DEIMv2-S	80.1	43.7	83.7	35.4	54.3	**31.6**
DMA-Det (ours)	79.5	45.3	**84.6**	41.2	**57.0**	31.1

TABLE VIII

DETECTION PERFORMANCE COMPARISON ON A SUBSET OF THE MASATI V2 DATASET. FOR DMA-DET, ONLY THE PROPOSED DETECTION NETWORK IS EVALUATED, WITHOUT DIFFUSION AUGMENTATION OR SEQF. BOLD AND UNDERLINE INDICATE THE BEST AND SECOND-BEST RESULTS IN EACH COLUMN, RESPECTIVELY

Method	AR	mAP_{50}	mAP_{50-95}
YOLOv10-L	72.3	77.3	35.9
YOLOv11-L	71.1	80.1	37.8
YOLOv12-L	71.6	81.3	38.4
YOLOv26-L	73.5	**83.6**	**39.7**
MS-DETR	90.7	69.4	26.8
HPR	91.6	83.4	37.4
S-DETR	86.3	75.7	31.4
R-DETR	85.2	71.8	32.0
RT-DETRv1-X	79.0	79.5	36.0
RT-DETRv2-X	91.1	82.7	37.2
PCViT	82.0	74.8	29.7
DEIMv2-S	92.6	82.9	36.9
DMA-Det (ours)	**93.3**	**83.6**	37.4

additional object detection dataset based on the GF-1 WFV imagery used in the semantic segmentation study of Liu et al. [27]. Since the data split in that study was designed for semantic segmentation and would lead to class imbalance in the present three-category detection task, we reprocess the imagery by cropping it into 1024 × 1024 patches and manually annotating the resulting patches with horizontal bounding boxes for red tide, ship, and island, yielding 1170 annotated instances in total. The resulting patches are randomly divided into training, validation, and test sets at a ratio of approximately 70%/15%/15%.

As presented in Table VII, DMA-Det achieves the highest mAP_{50} of 57.0%, exceeding the second-best result by 2.4 percentage points. Its mAP_{50-95} reaches 31.1%, tying for second place and trailing the best result by only 0.5 percentage points. At the category level, DMA-Det obtains the best island AP_{50} of 84.6% and the second-best red tide AP_{50} of 41.2%. Overall, these results suggest that DMA-Det can achieve competitive performance for multicategory detection on external remote sensing imagery.

5) Evaluation on the MASATI v2 Dataset: To further evaluate the applicability of the proposed detection network beyond NCSMRS, we conduct experiments on MASATI v2 [52], [53], a publicly available remote sensing dataset containing optical RGB imagery of various types of vessels in maritime scenes. We use 2368 ship-containing images from the *ship*, *multi*, and *coast & ship* scene categories. The *detail* category is excluded because vessels occupy a relatively large proportion of the image area, making these images less representative of the remote sensing detection setting considered. The *sea*, *coast*, and *land* categories are not included because they contain no annotated vessels. After merging the images from the three selected scene categories, the resulting dataset is randomly divided into training, validation, and test sets at a ratio of 70%/15%/15%.

To isolate the cross-dataset applicability of the proposed detection network, diffusion augmentation and SEQF are not applied in this experiment. As presented in Table VIII, DMA-Det (detector only) achieves the highest AR of 93.3% and ties with YOLOv26-L for the best mAP_{50} of 83.6%. However, its mAP_{50-95} of 37.4% is 2.3 percentage points below YOLOv26-L, suggesting room for improvement in precise localization under stricter IoU thresholds. Overall, these results indicate that the proposed detection network remains competi-

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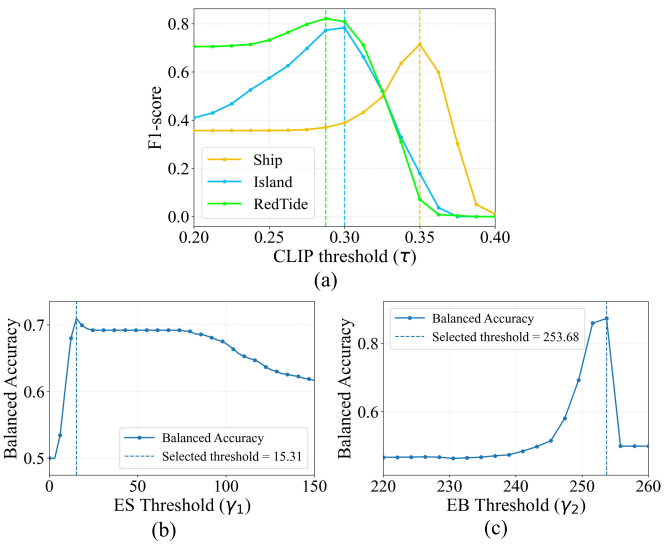


Fig. 6. SEQF threshold selection. (a) Per-category sweep of CLIP-based image–text similarity thresholds τ for SAV, with validation $F1$ scores reported. (b)–(c) Balanced accuracy as the edge sharpness threshold γ_1 and the edge-brightness threshold γ_2 are swept in EQA. Dashed vertical lines indicate the finalized threshold selections on each curve.

tive on an external public dataset for ship detection in maritime remote sensing imagery.

D. Parameter Analysis

Fig. 6 shows the threshold selection results for SAV and EQA. For SAV, the threshold of the CLIP-based image–text similarity score is swept separately for each category, and the category-specific threshold is selected as the value that maximizes the $F1$ score in semantic consistency verification. The $F1$ score varies noticeably across categories, indicating that a unified threshold is insufficient for semantic consistency filtering. Therefore, category-specific thresholds are adopted, with $\tau_{\text{ship}} = 0.35$, $\tau_{\text{island}} = 0.30$, and $\tau_{\text{red tide}} = 0.29$. For EQA, the generated samples are labeled as valid or invalid according to edge quality, and the edge filtering thresholds are selected by maximizing balanced accuracy for this binary classification task. The best performance is obtained when the edge sharpness threshold γ_1 is 15.31, and the edge-brightness threshold γ_2 is 253.68. These results highlight the importance of threshold selection, as inappropriate thresholds may reduce the reliability of the filtering process by either retaining low-quality samples or removing valid synthetic samples. The current threshold values are determined on the NCSMRS dataset. When SEQF is transferred to different regions or sensors, the filtering strategy itself remains applicable, but the specific numerical thresholds may need recalibration because target morphology, target scale, imaging conditions, and spatial resolution may vary.

Representative examples of synthetic samples filtered out by SEQF are shown in Fig. 7. Samples filtered out by SAV exhibit semantic inconsistency between the generated content and the category prompt, while samples filtered out by EQA contain boundary-quality defects such as blurred boundaries and unnatural halo artifacts. These examples qualitatively

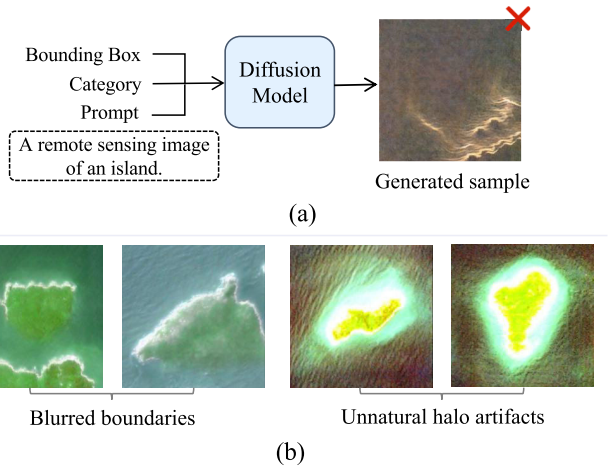


Fig. 7. Examples of synthetic samples filtered out by SEQF. (a) Semantic inconsistency identified by SAV. (b) Boundary-quality defects identified by EQA, including blurred boundaries and unnatural halo artifacts.

TABLE IX

ABLATION STUDY OF THE PROPOSED DMA-DET ON THE NCSMRS DATASET. SEQF IS APPLIED ONLY IF SYNTHETIC IMAGES ARE USED. BOLD AND UNDERLINE INDICATE THE BEST AND SECOND-BEST RESULTS IN EACH COLUMN, RESPECTIVELY

Data Augmentation		MDSA	AR	AP_{50}			mAP_{50}	$mAP_{50\text{-}95}$
Synth.	SEQF			Red Tide	Ship	Island		
-	-	-	**97.4**	53.8	79.1	87.2	73.4	47.7
✓	-	-	97.1	53.9	77.5	**90.5**	74.0	48.2
✓	✓	-	**97.4**	54.8	80.9	89.5	75.1	48.8
-	-	✓	97.1	54.4	82.3	88.0	74.9	49.0
✓	-	✓	97.0	52.7	80.0	87.2	73.3	47.4
✓	✓	✓	97.2	**55.8**	**83.9**	89.0	**76.2**	**49.7**

illustrate how SEQF helps improve the reliability of diffusion-generated training samples.

E. Ablation Studies

To validate the effectiveness of each component in DMA-Det, we conduct ablation experiments on the NCSMRS dataset, as presented in Table IX. The real-data-only baseline achieves 73.4% mAP_{50} and 47.7% $mAP_{50\text{-}95}$. Directly adding synthetic samples improves the two metrics to 74.0% and 48.2%, respectively, indicating that diffusion-based augmentation can provide useful additional supervision. After SEQF is applied, the performance further increases to 75.1% mAP_{50} and 48.8% $mAP_{50\text{-}95}$, showing that quality filtering improves the reliability of synthetic samples.

MDSA also provides clear performance gains when used with real data only. It improves mAP_{50} from 73.4% to 74.9% and $mAP_{50\text{-}95}$ from 47.7% to 49.0%, demonstrating its effectiveness in enhancing target-aware spatial representation and suppressing background interference from cloud cover and stripe noise. However, directly combining MDSA with unfiltered synthetic samples decreases the performance to 73.3% mAP_{50} and 47.4% $mAP_{50\text{-}95}$, indicating that low-

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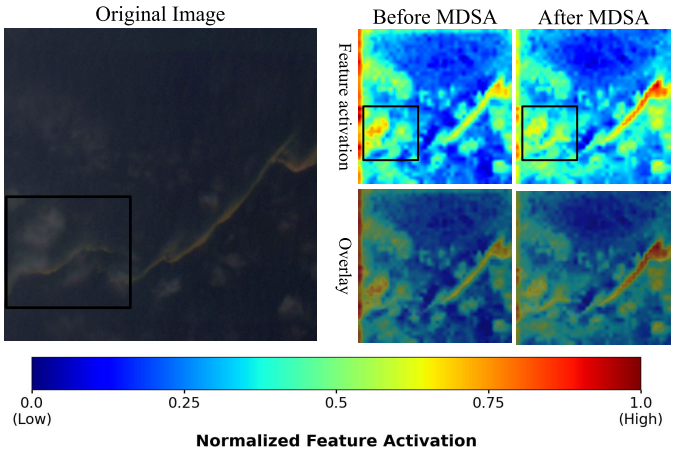


Fig. 8. Qualitative visualization of normalized backbone feature activations before and after applying MDSA. Without MDSA, feature activations tend to show relatively stronger responses in regions affected by cloud interference. With MDSA, the normalized activations become more concentrated in filamentary red-tide regions and less responsive to cloud-induced background patterns.

TABLE X

INTERNAL ABLATION STUDY OF THE SEQF MODULE. DIFFUSION AUGMENTATION AND MDSA ARE ENABLED FOR ALL SETTINGS. BOLD AND UNDERLINE INDICATE THE BEST AND SECOND-BEST RESULTS IN EACH COLUMN, RESPECTIVELY

SAV	EQA	AR	AP_{50}			mAP_{50}	$mAP_{50\text{-}95}$
			Red Tide	Ship	Island		
-	-	97.0	52.7	80.0	87.2	73.3	47.4
✓	-	96.6	<u>53.2</u>	<u>80.3</u>	<u>89.5</u>	74.3	<u>48.8</u>
-	✓	**97.3**	52.9	80.1	**90.7**	<u>74.6</u>	48.2
✓	✓	<u>97.2</u>	**55.8**	**83.9**	89.0	**76.2**	**49.7**

quality synthetic samples may introduce harmful noise. After SEQF filtering, the combination of diffusion augmentation and MDSA achieves the best results, reaching 76.2% mAP_{50} and 49.7% $mAP_{50\text{-}95}$. A visual comparison of normalized feature activations before and after applying MDSA is provided in Fig. 8, offering qualitative supporting evidence consistent with the feature recalibration effect of MDSA. In addition, MDSA introduces only 0.295 M additional parameters and 1.643 G additional FLOPs, with a slight decrease in inference speed from 123.26 FPS to 118.27 FPS.

The full DMA-Det integrates synthetic augmentation, SEQF, and MDSA, achieving the highest overall accuracy in terms of mAP_{50} and $mAP_{50\text{-}95}$. It also obtains the strongest red tide and ship AP_{50} scores, reaching 55.8% and 83.9%, respectively. Overall, synthetic augmentation alleviates sample scarcity, SEQF improves synthetic sample reliability, and MDSA enhances spatial feature modeling while suppressing background interference, jointly leading to robust detection performance.

Furthermore, to clarify the contribution of each component in SEQF, we evaluate SAV and EQA separately while keeping the diffusion augmentation pipeline and MDSA enabled. As presented in Table X, both components improve detection performance over the setting with unfiltered synthetic samples, indicating that semantic consistency validation and EQA both contribute to the reliability of augmented data. The full SEQF achieves the best mAP_{50} and $mAP_{50\text{-}95}$, suggesting that the two filtering components are complementary.

V. CONCLUSION

This article investigates marine object detection in complex nearshore scenes. To support this task, we construct the NCSMRS dataset, which covers three typical nearshore scene types, including cloud cover, stripe noise, and clean background, with bounding-box annotations for red tide, ships, and islands. We further propose DMA-Det, a detection framework that integrates layout-guided diffusion augmentation, SEQF, and MDSA. The diffusion model generates additional training samples to alleviate the shortage of difficult samples, SEQF improves synthetic data reliability by filtering samples based on semantic consistency and edge quality, and MDSA models axial and nonaxial spatial dependencies and then recalibrates multiscale features, thereby suppressing background interference from cloud cover and stripe noise while enhancing target-aware feature representations. Experimental results demonstrate that DMA-Det outperforms representative methods on the NCSMRS dataset and shows robust detection performance under cloud-cover and stripe-noise interference. Ablation studies confirm that diffusion augmentation, SEQF, and MDSA effectively alleviate sample scarcity, improve synthetic sample reliability, and suppress background interference, respectively.

However, DMA-Det still has several limitations. First, it mainly relies on RGB images, while multispectral information remains insufficiently explored. Second, missed detections may still occur in severely degraded scenes with thick clouds, strong stripe noise, or weak target responses. Third, the current NCSMRS dataset is collected from limited nearshore regions and time periods, and its generalization ability across different coastal areas, seasons, and sensors still requires further validation. Finally, bounding-box annotations inevitably include nontarget–background around diffuse red-tide regions. Future work will focus on exploiting multispectral cues, expanding the dataset coverage, improving detection robustness in severely degraded nearshore scenes, investigating oriented bounding-box detection to reduce background inclusion, and exploring box-guided local segmentation after detection to enable more accurate and comprehensive spatial characterization of red-tide regions.

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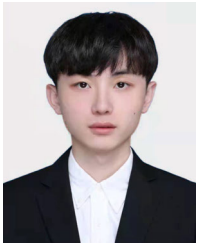
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